MCP Gateway Research: Dynamic Tool Discovery Solutions

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MCP Gateway Research: Dynamic Tool Discovery Solutions

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Executive Summary

This research investigates MCP (Model Context Protocol) gateways that specifically address **context pollution**—the problem where MCP servers flood agents with extensive tool schemas before any conversation begins. The ideal solution involves **dynamic tool discovery** where gateways present minimal tools upfront and use semantic routing to expose relevant tools on-demand.

Key Finding

The most promising approaches for your use case are:

1. **MCP-Zero Framework** - Academic research demonstrating 98% token reduction through hierarchical vector routing
2. **IBM Context Forge** - Open-source gateway with federation, semantic discovery, and admin UI
3. **Amazon Bedrock AgentCore Gateway** - Managed service with semantic tool selection
4. **Agentgateway (Linux Foundation)** - High-performance Rust implementation with built-in UI

The Context Pollution Problem

Current MCP implementations inject comprehensive tool schemas into system prompts, forcing agents into passive selection from pre-defined options. This creates massive context overhead:

- GitHub MCP server alone: **4,600+ tokens** for 26 tools
- Comprehensive ecosystems: **248,000+ tokens** before first turn
- Result: Degraded performance, slower responses, higher costs

The Solution: Dynamic Tool Discovery

The emerging best practice is to present **one or two meta-tools** that interface with a vector store of tool descriptions. The agent requests capabilities semantically, and only relevant tools are injected into context.

Detailed Gateway Analysis

Tier 1: Best Matches for Your Requirements

1. IBM MCP Context Forge

Repository: https://github.com/IBM/mcp-context-forge **Documentation:** https://ibm.github.io/mcp-context-forge/

Criteria	Assessment
Dynamic Tool Discovery	Yes - mDNS auto-discovery, health monitoring, capability merging
Context Pollution Solution	Federates and filters tools across gateways
Web UI	Yes - Admin UI (HTMX + Alpine.js) with real-time logs
Docker Support	Yes - Docker/Podman, Redis-backed caching
Streamable HTTP	Yes - HTTP, JSON-RPC, WebSocket, SSE, stdio, streamable-HTTP
Project Maturity	IBM community project, FastAPI-based
Unique Features	REST-to-MCP adaptation, A2A agent integration, OpenTelemetry

Key Advantage: Federation model allows multiple gateways to work together with automatic registry merging. This solves the local vs. remote MCP challenge by bridging environments transparently.

Best For: Complex deployments spanning local and cloud environments.

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2. Agentgateway (Linux Foundation)

Repository: https://github.com/agentgateway/agentgateway **Website:** https://agentgateway.dev/

Criteria	Assessment
Dynamic Tool Discovery	Yes - A2A and MCP protocol support for dynamic communication
Context Pollution Solution	RBAC-based tool filtering and access control
Web UI	Yes - Built-in UI for exploring agent-to-agent/agent-to-tool
Docker Support	Yes - Multiple Dockerfiles, Kubernetes via Gateway API
Streamable HTTP	Yes - Modern protocol support
Project Maturity	**Linux Foundation project**, 1.5k stars, 606 commits, 59 contributors
Unique Features	Rust-based (high performance), legacy API transformation (OpenAPI to MCP)

Key Advantage: Strong governance through Linux Foundation backing. Dynamic configuration updates via xDS without downtime. Can transform existing REST APIs into MCP resources.

Best For: Long-term stability, performance-critical deployments, enterprise adoption.

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3. Amazon Bedrock AgentCore Gateway

Documentation: https://aws.amazon.com/blogs/machine-learning/transform-your-mcp-architecture-unite-mcp-servers-through-agentcore-gateway/

Criteria	Assessment
Dynamic Tool Discovery	Yes - **Semantic Tool Selection** is core capability
Context Pollution Solution	Intelligent tool routing based on query semantics
Web UI	AWS Console integration
Docker Support	Managed service (not self-hosted)
Streamable HTTP	Full AWS integration
Project Maturity	**AWS managed service** - production-grade backing
Unique Features	Native MCP server target type, multi-server aggregation

Key Advantage: Purpose-built semantic tool selection. Can group multiple task-specific MCP servers behind single interface.

Limitation: Cloud-managed only, not self-hostable. Best as reference for desired behavior.

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4. MCP-Zero (Research Framework)

Paper: https://arxiv.org/html/2506.01056v3

Criteria	Assessment
Dynamic Tool Discovery	Yes - **Hierarchical Vector Routing** (two-stage coarse-to-fine)
Context Pollution Solution	**98% token reduction** demonstrated on APIBank
Web UI	Research framework, not production service
Docker Support	N/A
Unique Features	Iterative capability extension, MCP-tools dataset

Key Advantage: Academic validation of the semantic routing approach. Demonstrates that agents can actively request tools via structured format:

```
<tool_assistant>
  server: ...  # Platform/permission domain
  tool: ...    # Operation type + target
</tool_assistant>
```

Scoring Formula: `score = (s_server × s_tool) × max(s_server, s_tool)`

Best For: Understanding the theoretical foundation; implementing similar patterns in other gateways.

Tier 2: Strong Contenders

5. MetaMCP

Repository: https://github.com/metatool-ai/metamcp

Criteria	Assessment
Dynamic Tool Discovery	Middleware-based tool filtering
Context Pollution Solution	Namespace-level interception, tool overrides
Web UI	Yes - Inspector interface, inline tool editing
Docker Support	Yes - Docker Compose recommended
Project Maturity	1.8k stars, 759 commits, 32 releases (v2.4.22)
Unique Features	Profile/namespace management, OIDC enterprise SSO

Note: You mentioned using MetaMCP already. Its main limitation is lack of semantic/vector-based discovery—it relies on manual namespace configuration.

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6. Docker MCP Gateway

Repository: https://github.com/docker/mcp-gateway

Criteria	Assessment
Dynamic Tool Discovery	Real-time tool enumeration from active servers
Context Pollution Solution	Container isolation, consistent configs
Web UI	**No** - CLI only (`docker mcp`)
Docker Support	**Native** - Docker CLI plugin
Project Maturity	**Docker-backed**, 1.1k stars, 756 commits
Unique Features	Container-based isolation, Docker Desktop integration

Limitation: No web UI; CLI-only management. May not solve context pollution directly.

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7. MCPJungle

Repository: https://github.com/mcpjungle/MCPJungle

Criteria	Assessment
Dynamic Tool Discovery	Automatic tool cataloging from registered servers
Context Pollution Solution	**Tool Groups** - expose only subsets
Web UI	Registry server interface
Docker Support	Yes - dev and prod compose configs
Unique Features	Cherry-pick specific tools, OpenTelemetry metrics

Key Advantage: Tool Groups directly address context pollution by allowing selective tool exposure.

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8. Unla

Repository: https://github.com/AmoyLab/Unla

Criteria	Assessment
Dynamic Tool Discovery	Hot-reload via OS signals, HTTP, or Redis PubSub
Context Pollution Solution	Session isolation, multi-tenancy
Web UI	Yes - Management interface with built-in MCP chat
Docker Support	Yes - K8s/Helm support
Unique Features	Zero-code API conversion, persistent sessions

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9. Archestra

Repository: https://github.com/archestra-ai/archestra

Criteria	Assessment
Dynamic Tool Discovery	Private MCP registry with governance
Context Pollution Solution	**Dual LLM architecture** isolating dangerous responses
Web UI	Yes - ChatGPT-like interface
Docker Support	Yes - `docker pull archestra/platform:latest`
Unique Features	Non-probabilistic security against prompt injection via MCP

Key Advantage: Unique security model that isolates MCP tool responses to prevent data exfiltration.

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10. MCP Gateway Registry (Agentic Community)

Repository: https://github.com/agentic-community/mcp-gateway-registry

Criteria	Assessment
Dynamic Tool Discovery	**Semantic search** via embeddings (sentence-transformers, OpenAI, LiteLLM)
Context Pollution Solution	Fine-grained tool/method permissions
Web UI	Yes - Interactive Registry dashboard, Grafana observability
Docker Support	Yes - Pre-built images, Podman, ECS Fargate
Project Maturity	362 stars, 304 commits, 19 contributors
Unique Features	A2A agent registry, Keycloak/Cognito integration

Key Advantage: Semantic search directly addresses dynamic discovery.

Tier 3: Specialized/Partial Solutions

Semantic Router (vLLM)

Reference: https://joshuaberkowitz.us/blog/github-repos-8/semantic-tool-intelligence-vllm-context-aware-tool-router-solves-the-tool-overload-crisis-1736

Not a gateway but a routing layer that can be integrated: - Pre-computed embeddings for each tool - Cosine similarity matching against queries - Top-K tool selection above similarity threshold - **Results:** 99.1% token reduction (127,315 → 1,084 tokens), accuracy improved 13.62% → 43.13%

Portkey AI Gateway

Repository: https://github.com/Portkey-AI/gateway

High-maturity project (10.2k stars, 10B+ tokens processed daily) but focused on LLM routing rather than MCP-specific tool discovery.

Comparison Matrix

Gateway	Dynamic Discovery	Context Solution	Web UI	Docker	Backing	Stars
IBM Context Forge	Yes (mDNS)	Federation	Yes	Yes	IBM	-
Agentgateway	Yes (A2A/MCP)	RBAC	Yes	Yes	Linux Foundation	1.5k
AWS AgentCore	Yes (Semantic)	Native	Console	Managed	AWS	N/A
MCP-Zero	Yes (Vector)	98% reduction	No	No	Academic	N/A
MetaMCP	Partial	Namespaces	Yes	Yes	Community	1.8k
Docker MCP Gateway	Yes (Enum)	Isolation	No	Native	Docker	1.1k
MCPJungle	Yes	Tool Groups	Yes	Yes	Community	-
Unla	Yes (Hot-reload)	Sessions	Yes	Yes	Community	-
Archestra	Yes (Registry)	Dual LLM	Yes	Yes	Archestra AI	-
MCP Gateway Registry	Yes (Semantic)	Permissions	Yes	Yes	Community	362

Recommendation

Primary Choice: IBM Context Forge + Semantic Router Pattern

Rationale: 1. **Federation** solves your local/remote MCP bridging challenge 2. **Admin UI** meets your web management requirement 3. **REST-to-MCP adaptation** future-proofs API integration 4. **Open source** with IBM community backing (not a short-lived project) 5. **Streamable HTTP** supported (not deprecated SSE)

Enhancement: Integrate the semantic router pattern (from vLLM research or MCP-Zero approach) for true dynamic tool discovery. This would involve: - Vector store of tool descriptions - Single “discovery” tool exposed to agent - Semantic matching returns only relevant tools per query

Secondary Choice: Agentgateway

If Linux Foundation governance and Rust performance are priorities, Agentgateway offers: - Long-term project sustainability - Dynamic configuration without downtime - Legacy API transformation

Worth Watching: MCP Gateway Registry

The agentic-community project already implements semantic search with embeddings. Combined with its existing web UI and Docker support, it may evolve into the most complete solution.

Protocol Considerations

Streamable HTTP vs SSE

You correctly noted that SSE is deprecated. Current status: - **Streamable HTTP** is the emerging standard - IBM Context Forge, Agentgateway, and most Tier 1 solutions support it - MetaMCP supports both SSE and Streamable HTTP

Local-to-Remote File Transfer

This remains challenging. IBM Context Forge’s federation model is the closest solution, allowing local gateways to bridge to remote services. The pattern:

```
Local Files → Local MCP Gateway → Federation → Remote Gateway → Cloud Service
```

Implementation Path

1. **Deploy IBM Context Forge** via Docker with federation enabled
2. **Configure local MCP servers** (Home Assistant, local devices)
3. **Add remote MCPs** (Gemini transcription, File.AI, Context7)
4. **Implement semantic routing** by:
   - Creating tool description embeddings
   - Exposing single “discover_tools” endpoint

    - Using vector similarity for tool selection
5. **Connect Claude Code** to single gateway endpoint

Sources

- MCP-Zero Paper
- IBM Context Forge
- Agentgateway
- AWS AgentCore Gateway
- vLLM Semantic Router
- Awesome MCP Gateways
- Composio MCP Gateway Guide